Basics of Information Security (4360702)

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Diploma in Computer Engineering

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Chapter 5
Cyber Security

Topics To be Covered

- Introduction to Cyber Security
- Types of Cyber Attacks
- Vulnerabilities in Cyber Security
- Intruders
- Hackers
- Cyber Threats (Worms, Virus, Ad-Ware, Spy-Ware, Trojans, Covert Channels, Backdoors, Bots, IP Spoofing, ARP Spoofing, Session Hijacking).

- Cyber Crimes and Types of it
- Hacking
- Attack Vectors
- Cyberspace and Criminal Behavior
- Traditional Problems Associated with Computer/Cyber Crime.

Introduction to Cyber Security

- Cybersecurity is the practice of protecting computer systems, networks, and digital information from unauthorized access, attacks, damage, or theft.
- In an increasingly interconnected and digitized world, where individuals, organizations, and governments rely heavily on technology, cybersecurity plays a essential role in safeguarding sensitive data and ensuring the integrity and availability of digital assets.
- The rapid evolution of technology and the universal use of the internet have brought numerous benefits but have also introduced new challenges and risks.
- Cyber threats, including malware, phishing attacks, ransomware, and various other forms of cybercrime, have become more sophisticated and prevalent.
- As a result, the field of cybersecurity has emerged as a critical discipline aimed at understanding, preventing, and responding to these evolving threats.
- Key Components or basic principals of cybersecurity are listed below that has to maintain and don’t break during communication.
    - Confidentiality
    - Integrity
    - Availability
    - Authentication
    - Authorization
    - Non Repudiation

Types of Cyber Attacks

- Malware Attacks
  - Viruses: Malicious code that attaches itself to legitimate programs and spreads when those programs are executed.
  - Worms: Self-replicating malware that spreads across networks without user intervention.
  - Trojans: Malware disguised as legitimate software, often tricking users into installing it.
- Phishing Attacks
  - Email Phishing: Deceptive emails that aim to trick recipients into revealing sensitive information or clicking on malicious links.
- Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks
  - DoS Attacks: Overloading a system or network to make it unavailable to users.
  - DDoS Attacks: Coordinating a large number of compromised devices to flood a target with traffic, causing a service disruption.
- Man-in-the-Middle (MitM) Attacks
  - Eavesdropping: Intercepting and monitoring communication between two parties without their knowledge.
  - Session Hijacking: Taking over an established session between a user and a system.

- SQL Injection
  - Exploiting vulnerabilities in web applications to inject malicious SQL code, enabling unauthorized access or manipulation of databases.
- Cross-Site Scripting (XSS)
  - Injecting malicious scripts into web pages that are viewed by other users, potentially leading to the theft of sensitive information.
- Ransomware Attacks
  - Encrypting a user's data and demanding a ransom payment for its release. Ransomware can target individuals, businesses, or even entire networks.
- Social Engineering Attacks
  - Manipulating individuals into divulging sensitive information through psychological manipulation or deception in connected devices to gain unauthorized access or control.
- Insider Threats
  - Attacks or security breaches initiated by individuals within an organization, either intentionally or unintentionally.

Vulnerabilities in Cyber Security

- Vulnerabilities in cybersecurity refer to weaknesses in computer systems, networks, applications, and processes that could be exploited by attackers to compromise the security of information or systems.
- Here are some common categories of vulnerabilities:
  - ❑ Software Vulnerabilities
    - Unpatched Software: Failure to apply security patches and updates can leave systems exposed to known vulnerabilities.
    - Insecure Configurations: Incorrectly configured software or systems may have settings that can be exploited by attackers.
  - ❑ Network Vulnerabilities
    - Weak Network Security: Inadequate firewall configurations, lack of intrusion detection/prevention systems, and weak network segmentation can expose networks to attacks.
    - Open Ports and Services: Unnecessary open ports and services increase the attack surface and may lead to unauthorized access.
  - ❑ Human Factor Vulnerabilities
    - Weak Passwords: Poor password practices, such as using easily guessable passwords or not changing default credentials, can lead to unauthorized access.

- Physical Security Vulnerabilities
  - Unauthorized Access: Lack of physical access controls, such as unsecured server rooms or unlocked devices, can result in unauthorized access to critical systems.
- Application-Level Vulnerabilities
  - Code Injection: Poorly designed applications may be susceptible to injection attacks, such as SQL injection or cross-site scripting (XSS).
  - Insecure APIs: Flaws in the design or implementation of application programming interfaces (APIs) can lead to data breaches and unauthorized access.
- Operating System Vulnerabilities
  - Outdated OS: Using an outdated or unsupported operating system may expose systems to known vulnerabilities that have not been patched.
  - Insufficient Access Controls: Inadequate permissions and access controls can result in unauthorized users gaining escalated privileges.
- Supply Chain Vulnerabilities
  - Third-Party Software: Dependencies on third-party software or libraries with vulnerabilities can introduce security risks.
  - Compromised Hardware: Malicious actors may tamper with hardware components during the supply chain process, compromising the integrity of systems.

Intruders

- The most common threat to security is an attack by an Intruder.
- Intruders are often referred to as hackers and are the most harmful factors contributing to the vulnerability of security, they have immense knowledge and an in-depth understanding of technology and security.
- Intruders crack the privacy of users and aim to steal the confidential information of the users, the stolen information is then sold to third parties, which aim at misusing the information for their own personal or professional gains.
- Three Classes of Intruders:
  - Masquerader
    - An individual who is not authorized to use the computer and who penetrates a system's access controls to exploit a legitimate user's account.
  - Misfeasor
    - A legitimate user who accesses data, programs, or resources for which such access is not authorized, or who is authorized for such access but misuses his or her privileges.
  - Clandestine User
    - An individual who seizes supervisory control of the system and uses this control to escape auditing and access controls.

Hackers

- Hacker can be used to describe individuals with various skills and motivations related to computer systems and networks, It's important to note that not all hackers engage in malicious activities.
- Here are some common types of hackers,
  - ❑ White Hat Hackers (Ethical Hackers)
    - It uses their skills to identify and fix security vulnerabilities.
    - They work to strengthen the security of systems, often employed by organizations to conduct penetration testing and security assessments.
  - ❑ Black Hat Hackers
    - Black hat hackers engage in malicious activities for personal gain or with harmful intent.
    - They may exploit vulnerabilities, steal sensitive information, launch attacks, or cause damage to computer systems. These individuals are typically associated with illegal and unethical activities.
  - ❑ Gray Hat Hackers
    - Gray hat hackers fall somewhere between white hat and black hat hackers.
    - They may break into systems without explicit permission, but they claim to do so for ethical reasons, such as exposing vulnerabilities or flaws. However, their actions may still be considered illegal.

- Hacktivists
  - Hacktivists are individuals or groups who hack for political, social, or ideological reasons. Their activities often involve defacing websites, disrupting services, or stealing and leaking information to promote a specific cause.
- Cyber Criminals
  - Cyber criminals are individuals or groups motivated by financial gain. They may engage in activities such as stealing personal information, conducting ransomware attacks, or committing fraud for monetary benefits.
- State Sponsored Hackers
  - State-sponsored hackers are individuals or groups supported by governments to conduct cyber warfare, or other activities in the interest of the sponsoring nation.
- Malware Authors
  - Individuals who create and distribute malicious software, such as viruses, worms, Trojans, and ransomware

Cyber Threats

- Cyber threats refer to malicious activities or risks that target computer systems, networks, and digital infrastructure with the intent of causing harm.
- These threats can compromise the confidentiality, integrity, or availability of information, leading to financial losses, reputation damage, or disruption of essential services.
- Here are some common types of cyber threats:
  - Worms
    - Worms are self-replicating malware that can spread across networks without user intervention.
    - They typically exploit vulnerabilities in operating systems or software to infect and spread.
  - Virus
    - Viruses are malicious programs that attach themselves to legitimate files or software.
    - They replicate when the infected program runs, spreading to other files and systems.
    - Viruses can cause various types of damage, such as data corruption or deletion.

- Adware
  - Adware, short for advertising-supported software, displays unwanted advertisements on a user's device. While not always malicious, some adware can compromise user privacy by collecting and transmitting personal information.
- Spyware
  - Spyware is designed to secretly monitor and collect user information without their knowledge.
  - It can capture keystrokes, browser history, login credentials, and other sensitive data, often for malicious purposes.
- Trojans
  - Trojans, or Trojan horses, are deceptive programs that appear legitimate but contain malicious code.
  - They can perform various harmful actions, such as granting unauthorized access, stealing data, or facilitating other types of malwares.
- Covert Channels
  - Covert channels are methods of communication that are hidden or not easily detectable, In a security context, they can be used by attackers to transmit information stealthily, bypassing security measures.

- Backdoors
  - Backdoors are hidden access points in a system that allow unauthorized entry and control.
  - They are often created by attackers to maintain persistent access to compromised systems.
- Bots
  - Bots are networks of compromised computers controlled by a single entity for activities like launching DDoS Attacks or spreading malware.
- IP Spoofing
  - IP spoofing involves manipulating the source IP address in network packets to deceive recipients about the origin of the message.
  - Attackers may use IP spoofing to bypass access controls or launch attacks while appearing to be from a trusted source.
- ARP Spoofing
  - ARP (Address Resolution Protocol) spoofing involves linking a false MAC address with an IP address within a network. This can lead to traffic interception, eavesdropping, or man-in-the-middle attacks.
- Session Hijacking
  - Session hijacking occurs when an attacker intercepts and takes control of an active user session. This can lead to unauthorized access to sensitive information or systems.

Cyber Crime and Its Types

- Cybercrime refers to criminal activities carried out using the Internet, computer systems, or other digital technologies.
- These offenses may target individuals, organizations, or governments, and they can encompass a wide range of illegal activities.
- Here are some common types of cyber crime:
  - Hacking
    - Unauthorized access to computer systems or networks to gather, manipulate, or disrupt data.
  - Malware
    - The creation and dissemination of malicious software (malware), such as viruses, worms, trojan horses, and ransomware, with the intent to harm or gain unauthorized access.
  - Phishing
    - Deceptive attempts to acquire sensitive information, such as usernames, passwords, and financial details, by posing as a trustworthy entity in electronic communication.
  - Identity Theft
    - Unauthorized use of someone else's personal information to commit fraud or other criminal activities.

- Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks
  - Overloading a target's computer system or network to make it unavailable to users.
- Online Fraud
  - Various forms of fraud conducted over the Internet, including online scams, fake auctions, and fraudulent online purchases.
- Child Exploitation
  - The use of digital technology to exploit children, including online grooming, child pornography, and other forms of abuse.
- Cyberbullying
  - Harassment, pressure, or mistreatment that occurs online, often through social media platforms, email, or messaging apps.
- Financial Cybercrimes
  - Crimes involving the theft of financial information, such as credit card fraud, online banking fraud, and cryptocurrency-related offenses.
- Cyber Extortion
  - Threatening to reveal sensitive information, launch attacks, or damage systems unless a ransom is paid.

- Insider Threats
  - Unauthorized or malicious activities conducted by individuals within an organization, such as employees or contractors, who misuse their access to compromise security.
- Rogue Software
  - Distributing fake or malicious software that misleads users into thinking it is legitimate or useful, often leading to financial losses or security breaches.
- Cyber Stalking
  - Repeated, unwanted, and intrusive online behaviors, including monitoring, harassment, or threats, directed towards an individual or group.

Hacking

- Hacking refers to the unauthorized access, manipulation, or exploitation of computer systems, networks, or electronic devices.
- Individuals who engage in hacking activities are commonly referred to as "Hackers"
- Here are some key aspects of Hacking:
  - Unauthorized Access
    - Hacking often involves gaining access to computer systems or networks without the explicit permission of the system owner or administrator.
  - Types of Hackers
    - White Hat Hackers (Ethical Hackers)
      - It uses their skills to identify and fix security vulnerabilities.
      - They work to strengthen the security of systems, often employed by organizations to conduct penetration testing and security assessments.
    - Black Hat Hackers
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- Malware Authors
  - Individuals who create and distribute malicious software, such as viruses, worms, Trojans, and ransomware

Method of Hacking

- Social Engineering: Manipulating individuals into divulging confidential information through deceptive means.
- Phishing: Using fake websites, emails, or messages to trick individuals into providing sensitive information.
- Brute Force Attacks: Attempting to gain access by systematically trying all possible combinations of passwords or encryption keys.
- Exploiting Software Vulnerabilities: Identifying and taking advantage of weaknesses in software or applications.
- Man-in-the-Middle Attacks: Intercepting and altering communication between two parties without their knowledge.

Motivations for Hacking

- Financial Gain: Stealing financial information, conducting online fraud, or demanding ransom.
- Spying: Gathering sensitive information for political, economic, or military purposes.
- Hacktivism: Hacking for ideological or political reasons to promote a social or political agenda.
- Personal Satisfaction or Challenge: Some hackers may be motivated by the intellectual challenge of breaking into systems.

Attack Vectors

- Attack Vectors are paths or methods that cyber attackers use to gain unauthorized access to computer systems, networks, or devices.
- These vectors represent the various paths through which malicious actors exploit vulnerabilities to compromise the security of a system.
- Here are some common Attack Vectors:
  - Phishing
    - Attackers send deceptive emails, often disguised as legitimate communications, to trick individuals into revealing sensitive information or clicking on malicious links.
  - Malware
    - Email Attachments: Malicious software can be delivered through email attachments that, when opened, infect the user's system.
    - Infected Websites: Drive-by downloads occur when users visit compromised websites, unknowingly downloading malware onto their systems.
  - Social Engineering
    - Manipulation of Trust: Attackers exploit human psychology to manipulate individuals into divulging sensitive information or taking specific actions.

- Brute Force Attacks
  - Attempting to gain unauthorized access to a system by systematically trying all possible combinations of passwords or encryption keys.
- SQL Injection
  - Exploiting vulnerabilities in web applications to inject malicious SQL code into input fields, potentially allowing unauthorized access to databases.
- Cross-Site Scripting (XSS)
  - Injecting malicious scripts into websites, which are then executed by users' browsers, allowing attackers to steal information or perform actions on behalf of the user.
- Man-in-the-Middle (MitM) Attacks
  - Intercepting and potentially altering communication between two parties, allowing attackers to eavesdrop on sensitive information.
- Physical Access
  - Gaining access to physical devices or infrastructure to compromise security, install malware, or steal sensitive information.
- USB and Removable Media Attacks
  - Distributing malware or executing malicious actions through infected USB drives or other removable media.
- Bluetooth and IoT Exploitation
  - Exploiting vulnerabilities in Bluetooth connections or Internet of Things (IoT) devices to gain unauthorized access.

Cyberspace and Criminal Behavior

- Cyberspace, the virtual environment created by interconnected computer systems and networks, has provided a platform for numerous activities, including both legal and criminal behaviors.
- Criminals influence the secrecy using various illegal activities.
- Here are some ways in which cyberspace intersects with criminal behavior:
    - Cybercrime
    - Cyber Spying
    - Cyber Terrorism
    - Illegal Online Markets
    - Child Mistreatment
    - Cyberbullying
    - Financial Crimes
    - Unauthorized Access and Data Theft
    - Online Scams
    - Disinformation and Cyber Influence Operations
- Addressing criminal behaviour in cyberspace requires international cooperation, robust cybersecurity measures, regulation, and law enforcement efforts.
- Governments, organizations, and individuals need to stay attentive, adopt security best practices, and work together to combat the evolving landscape of cybercrime.

Traditional Problems Associated with Computer Crime

- Jurisdictional Challenges
  - Determining jurisdiction can be difficult when cybercrimes transcend national borders.
  - Crimes committed online may involve perpetrators and victims in different countries, making it challenging for law enforcement to pursue and prosecute offenders.
- Anonymity and Pseudonymity
  - Cybercriminals often use various techniques to conceal their identities, such as operating under aliases or using anonymizing tools like virtual private networks (VPNs),This anonymity makes it challenging for law enforcement to trace and apprehend offenders.
- Rapid Evolution of Technology
  - Technology evolves quickly, and cybercriminals adapt their methods accordingly.
  - Law enforcement agencies and security measures may struggle to keep pace with the constantly changing landscape of cyber threats.
- Attribution Difficulties
  - Determining the true identity of cybercriminals and attributing specific cybercrimes to individuals or groups can be challenging.

- Cross-Border Collaboration
  - Legal systems and law enforcement agencies across different countries often face challenges in collaborating effectively.
  - Varying laws, cultural differences, and bureaucratic hurdles can impede international efforts to combat cybercrime.
- Resource Limitations
  - Law enforcement agencies may lack the resources, both in terms of personnel and technology, to effectively investigate and combat the increasing volume and complexity of cybercrimes.
- Reporting and Awareness
  - Underreporting of cybercrimes is common due to victims' unwillingness to report incidents, lack of awareness, or fear of reputational damage.
- Evidence Collection and Preservation
  - Digital evidence, such as logs, files, and network data, may be easily manipulated or destroyed.
  - Properly collecting and preserving digital evidence for legal proceedings can be challenging, requiring specialized knowledge and tools.
- Public Perception and Education
  - Lack of awareness and understanding of cyber threats can contribute to individuals and organizations falling victim to cybercrimes.
  - Public education about online risks and best practices for cybersecurity is essential.

Important Questions

Unit-5 (CO-5)	
Sr No.	**Question**
1	Define Cyber Security.
2	List Cyber Attacks and Explain any Four. OR List types of cyber attacks.
3	List Vulnerabilities in Cyber Security.
4	Define Intruders,Explain its Types.
5	Define Hackers, Explain Its Types.
6	List Cyber Threats and Explain any Four. OR Discuss different types of threats.
7	Define Cybercrime and Explain its types in detail. OR List types of cybercrime.
8	Define Hacking and List Methods of Hacking.
9	List Attack Vectors and Explain any Four.
10	List Different ways by which cyberspace intersects with criminal behavior.
11	Explain Traditional Problems Associated with Computer/Cyber Crime.
12	Define worms, viruses and trojans
13	Describe cybercrime with its types in detail
14	Describe IP Spoofing, ARP spoofing and Session Hijacking in detail.